BRAIN 12

Esketamine (Spravato)

S form of ketamine, an NMDA receptor blocker, given as a nasal spray (Spravato)



CLASS NMDA receptor antagonist

STUDIED FOR Treatment-resistant depression

HUMAN EVIDENCE Large trials; FDA approved

US STATUS FDA approved (Spravato) · Schedule III · REMS

STUDIED FOR

- Resistant depression — HUMAN
- Relapse prevention — HUMAN
- Depression, alone — HUMAN
- Suicidal thoughts — HUMAN
- vs quetiapine — HUMAN

Mixed Phase 3 results. Some maker-run trials met their main goal and some did not. The label says it has not been shown to prevent suicide.

HOW IT WORKS IN THE BODY

Esketamine ▶ NMDA receptor ▶ Glutamate signals ▶ Depression

- **Label:** blocks the NMDA receptor, a receptor for glutamate
- **Label:** how it eases depression is unknown
- Used in a **certified health care setting**, with at least 2 hours of monitoring

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Phase 3 trials: TRANSFORM, SUSTAIN, ASPIRE
- FDA approved 2019; added uses 2020, 2025
- Two TRANSFORM trials missed their main goal

NOT YET ESTABLISHED

- Suicide prevention (label)
- Use in children (not approved)

QUICK PRACTICAL INFO

HALF-LIFE Terminal half-life 7 to 12 hours (label)

ROUTES STUDIED Nasal spray, in a certified health care setting (label)

PHARMACOKINETICS Peak 20 to 40 min; CYP2B6 and CYP3A4; about 48% bioavailable

SOLD AS Prescription nasal spray (Spravato) via REMS; Schedule III

MONITORED IN TRIALS Sedation, dissociation, breathing, blood pressure for 2 h+

SPORT (WADA) Not on the 2026 list

REGULATORY

FDA	Approved (Spravato, 2019); boxed warning; REMS
DEA	Schedule III (ketamine isomers)
EMA	Authorised 2019, with an SSRI or SNRI
WADA	Not listed

REFERENCES

1. FDA (DailyMed), Spravato prescribing information, Sep 2026
2. JAMA Psychiatry, SUSTAIN-1 relapse prevention trial, Sep 2019
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

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